

An Integrated Framework for Assessing Renewable Energy Impact on Frequency Stability: Case Study for Vietnam in 2030

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Abstract— Currently, the world is experiencing a significant rise in the share of Renewable Energy Sources (RES) in power systems. This growth helps reduce CO_2 emissions and lowers the overall generation costs of the power system. However, it also results in a decreased share of traditional generators, which are the main sources of system inertia. Therefore, a high penetration of RES leads to a reduction in system inertia, posing significant challenges to frequency stability. This paper proposes an integrated process to assess the impact of RES on frequency stability according to Vietnam's National Power Development Plan by 2030, when RES is predicted to participate in 54.4% of the total capacity of this system. The process begins with formulating and solving the Unit Commitment problem to create an optimal schedule that closely approximates actual operation. After that, dynamic stability was simulated in severe disturbances to assess the frequency indicators, including Inertia, Frequency Nadir (FN), and Rate of Change of Frequency (RoCoF). The analysis of these metrics is associated with evaluating changes in the operational plan when the rate of RES increases, providing direct insight into the impact of each scenario on frequency stability. The calculation and simulation processes were conducted using the PyPSA library with CPLEX solver and PSS/E.

Keywords—Renewable Energy Sources, Frequency Stability, Inertia, Frequency Nadir, Rate of Change of Frequency

I. INTRODUCTION

Currently, RES accounts for an increasingly high proportion of the world's installed power system capacity. Although this growth helps minimize the environmental impact and lower the cost of electricity production for the system [1], it also poses difficulties and challenges for system operation, particularly the issue of frequency stability.

Synchronous generators play an important role in maintaining frequency stability. The inertia of these synchronous generators slows down the dynamic responses in the system, facilitating frequency regulation when the system is subjected to disturbances [2]. However, the increasing penetration of RES, such as wind or solar power, has reduced the system inertia because RES are connected to the grid via power electronic converters and do not provide inertia. This problem could reduce the system's frequency response ability [3], [4]. This inertia reduction leads to faster frequency change rates, an increased Rate of Change of Frequency (RoCoF), and a low Frequency Nadir (FN), which results in frequency instability when the system is subjected to large disturbances

[5]. According to Vietnam's National Power Development Plan VIII (PDP8) for the 2021-2030 period, the total installed capacity of all generators is expected to reach between 183,291 MW and 236,363 MW by 2030. This system is designed to accommodate a predicted peak demand of 99,934 MW [6]. Notably, the installed capacity of RES is expected to range from 78,525 MW to 128,477 MW, equivalent to 42.84% to 54.35% of the total installed capacity [6]. Therefore, there is an urgent need to assess the frequency stability of Vietnam's power system in many different operating scenarios and propose solutions to improve frequency stability when the proportion of RES is high.

In recent years, several studies have focused on analyzing the impact of huge RES participation on system inertia and frequency stability. In study [7] in 2023, Farmer et al. pointed out that low inertia causes RoCoF to increase rapidly, posing a risk of frequency instability, while study [8] in 2022 on the Indian power system with a RES proportion up to 80% also recorded RoCoF exceeding 1 Hz/s during major faults and emphasized the urgent need for technical and operational solutions. The impact of very high PV penetration on the frequency response of three major US grids was also analyzed in [9]. Paper [10] has also investigated frequency stability when the proportion of RES increases and proposed a solution to integrate frequency stability constraints directly into the UC problem. This approach aims to achieve a balance between ensuring frequency stability and minimizing production costs for the system. In addition, paper [11] also proposed a method for quantitatively assessing power system frequency stability when the RES participation is high, based on the Input-to-State Stability (ISS) theory and the System Frequency Response (SFR) model. However, these studies only evaluated some fixed scenarios corresponding to a certain level of RES penetration, without comprehensively considering all hourly operating scenarios in the Unit Commitment (UC) optimization problem, which more closer to actual conditions. This significantly reduces practical applicability.

Since there are several issues identified, we propose an integrated process to assess the impact of RES on the frequency stability of the Vietnamese power system according to PDP8 until 2030. The process starts with solving the UC problem to obtain realistic system operating scenarios, then building dynamic models and system data for each scenario for computational simulations running to make assessments with the System Inertia, FN and RoCoF parameters. The

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calculation and simulation process is performed based on the PyPSA tool and PSS/E software. The impacts of RES on frequency stability indicators were analyzed, closely following the features in the operating scenarios in the Vietnam power system until 2030, including generator structure and load growth rate according to PDP8.

II. PROBLEM FORMULATION

A. Integrated Framework for Assessing the Impact of RES on Frequency Stability

In this paper, an integrated process is proposed to assess impacts of RES on frequency stability and implement calculation directly with Vietnamese power system according to Power Development Plan VIII by 2030 (Fig.1). It starts with collecting and processing data on the Vietnamese power system as input data for the original UC problem and the structure of the Vietnamese power grid in the PSS/E software. Then, build a mathematical model and solve the UC problem to obtain the results. Using those results and the system's initialization data on PSS/E to create hourly scenarios. The dynamic model was created to define all components in the system. After having these model, dynamic stability simulation was conducted with the newly created scenarios in severe system failures. Finally, the results of Inertia, FN, RoCoF were calculated and evaluated.

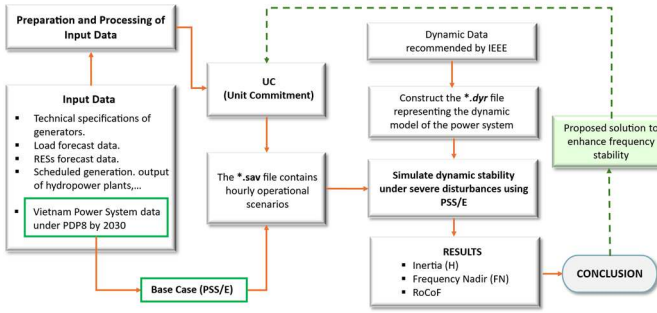


Fig. 1. Proposed Integrated Framework.

The process of building a dynamic model of the system is described in Fig.2. It begins with implementing a dynamic model for generators based on parameters from the IEEE and WECC templates in PSS/E. The model for traditional generators includes: generator models (GENROU, GENSAL), excitation system models (IEEE1, EXDC2, EXST1, SCR, ...), governor models (HYGOV, TGOV1, GAST). For wind turbines, the model REGCA1, electrical control model REECA1 and supplementary controllers such as REPCA1, WDTA1, WTPTA1, WTARA1 were utilized. For photovoltaic generators (PV), dynamic models PVGU1, PVEU1, PANELU1, IRRADU1 were used. After the model building process is completed, the initial steady state was initialized. This step is critical as the incorrect dynamic parameters might cause oscillation during initialization and make the subsequent dynamic simulation results inaccurate. Therefore, during the initialization process, if these are suspects, the tool presented in Fig.2 would find the generators with inaccurate dynamic parameters. Once all components are tuned, the system reaches a valid steady state condition and is ready for fault scenario simulations, such as sudden loss of an operating generator unit.

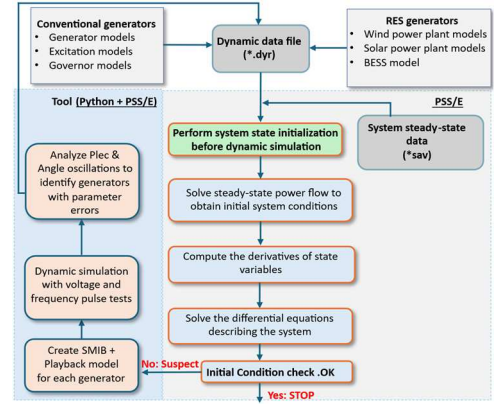


Fig. 2. Create a dynamic model of the power system.

B. Optimal Operation Scheduling

1) Objective Function

The objective function of the UC problem is to minimize the purchasing electricity cost.

$$\text{Min} \left(\sum_{t=1}^T (\sum_{i \in CV} u_{i,t} \cdot C_{i,t} \cdot P_{i,t} + y_{i,t} \cdot C_{i,t}^{ST} + z_{i,t} \cdot C_{i,t}^{SD}) \right) \quad (1)$$

Where CV is the set of synchronous generators. $u_{i,t}$, $y_{i,t}$, $z_{i,t}$ are binary variables representing ON/OFF, starting or shut down status of generator units, respectively. $C_{i,t}$, $C_{i,t}^{ST}$, $C_{i,t}^{SD}$ are the marginal cost, starting cost and shutdown cost of each unit.

2) Constraints

• Power balancing constraint

The total output from generators and storage systems at each time must be balanced with the demand from the load and the charging/pumping power of the storage systems.

$$\sum_g P_{g,t} + \sum_e (P_{e,t}^{BESSdisch} - P_{e,t}^{BESSch}) + \sum_p (P_{p,t}^{PHSdisch} - P_{p,t}^{PHSsch}) = P_{load,t}, \forall t \quad (2)$$

• Transmission line constraint

Power flow across all transmission lines must be below the maximum limits in both directions (3).

$$-P_l^{max} \leq P_{l,t} \leq P_l^{max}, \forall t \quad (3)$$

• Traditional generator constraints

The constraints for traditional generators in the UC problem are defined in equations (4) – (11). Among these, constraints (7)–(8) represent the minimum up/down time requirements of generating units. Equations (9) and (10) ensure that certain units with specific roles must always be operational (Must Run) or remain offline due to maintenance (Must Off) during the scheduling period. Additionally, equation (11) guarantees that the total energy output of hydropower plants does not exceed the allowable limit throughout the planning period, where NP represents the number of units in a hydropower plant.

$$u_{i,t} \cdot P_i^{min} \leq P_{i,t} \leq u_{i,t} \cdot P_i^{max}, \forall t \quad (4)$$

$$P_{i,t} - P_{i,t-1} \leq RU_i \cdot u_{i,t-1} \quad (5)$$

$$P_{i,t-1} - P_{i,t} \leq RD_i \cdot u_{i,t} \quad (6)$$

$$\sum_{t'=t}^{t+TU_i} u_{i,t'} \geq TU_i \cdot (u_{i,t} - u_{i,t-1}) \quad (7)$$

$$\sum_{t'=t}^{t+TD_i} (1 - u_{i,t'}) \geq TD_i \cdot (u_{i,t-1} - u_{i,t-1}) \quad (8)$$

$$u_{i,t} = 1 \text{ với } i \in MR, t \in [1, T] \quad (9)$$

$$u_{i,t} = 0 \text{ với } i \in MO, j \in T. \quad (10)$$

$$\sum_{hg=1}^{N_p} \sum_{t=1}^T u_{hg,t} \cdot P_{hg,t} \cdot \Delta t \leq A_p^{schedule} \quad (11)$$

- Spinning reserve constraints

Each area in the power system must ensure a sufficient amount of spinning reserve to contribute to frequency control, as expressed in equation (12). Where, R_t^{AREA} represents the required reserve capacity for each area.

$$\sum_{i'}^{n_t^{AREA}} (u_{i',t} \cdot P_{i'}^{max} - P_{i',t}) \geq R_t^{AREA}, \forall t \quad (12)$$

- Constraints for RES generators.

The output power of wind and solar generators must comply with constraints (13) and (14). Where, $P_{w,t}^{max}, P_{s,t}^{max}$ denote the available power output of wind and solar resources at each time step t , respectively.

$$0 \leq u_{w,t} \cdot P_{w,t} \leq u_{w,t} \cdot P_{w,t}^{max} \quad (13)$$

$$0 \leq u_{s,t} \cdot P_{s,t} \leq u_{s,t} \cdot P_{s,t}^{max} \quad (14)$$

- BESS Constraints.

The operation of the Battery Energy Storage System (BESS) must comply with constraints (15) – (19). Constraints (15) – (16) ensure the charging and discharging power limits of the BESS, while constraints (17) – (19) describe the energy charging/discharging process. Where, $\chi_{e,t}$ is a binary variable equal to 1 when the BESS is discharging and 0 when it is charging and η_e^{BESS} represents the charging/discharging efficiency of the BESS.

$$0 \leq P_{e,t}^{BESSdisch} \leq \chi_{e,t} \cdot P_e^{BESSmax} \quad (15)$$

$$0 \leq P_{e,t}^{BESSsch} \leq (1 - \chi_{e,t}) \cdot P_e^{BESSmax} \quad (16)$$

$$E_{e,t}^{BESS} = E_{e,t-1}^{BESS} + P_{e,t}^{BESSsch} \cdot \eta_e^{BESS} - \frac{P_{e,t}^{BESSdisch}}{\eta_e^{BESS}} \quad (17)$$

$$E_{e,t=0}^{BESS} = E_{e,t=T}^{BESS} \quad (18)$$

$$0.1 \times E_{e,rate}^{BESS} \leq E_{e,j}^{BESS} \leq 0.9 \times E_{e,rate}^{BESS} \quad (19)$$

- PSH Constraints.

The constraints for Pumped Storage Hydropower units are defined in equations (20) – (24). Similar to BESS, PSH units must also satisfy charging and discharging power limits, as expressed in (20) – (21). Where, $v_{p,t}$ is a binary variable indicating the pumping/generating state of the PSH units. The relationship between the reservoir energy level and the pumping/generating process is described by constraints (22) – (24). In this equation, $p'max$ denotes the number of units in the PSH plant p' .

$$0 \leq P_{p,t}^{PHSdisch} \leq v_{p,t} \cdot P_p^{PHSmax} \quad (20)$$

$$0 \leq P_{p,t}^{PHSsch} \leq (1 - v_{p,t}) \cdot P_p^{PHSmax} \quad (21)$$

$$E_{p',t}^{PHS} = E_{p',t-1}^{PHS} + \sum_{p=p_1}^{p'max} \left(P_{p,t}^{PHSsch} \cdot \eta_p^{PHSsch} - \frac{P_{p,t}^{PHSdisch}}{\eta_p^{PHSdisch}} \right) \quad (22)$$

$$E_{p',t=0}^{PHS} = E_{p',t=T}^{PHS} \quad (23)$$

$$0.1 \times E_{p',rate}^{PHS} \leq E_{p',t}^{PHS} \leq 0.9 \times E_{p',rate}^{PHS} \quad (24)$$

C. Frequency Stability.

- System Inertia.

When a power imbalance occurs between generation and load, the rate of change of frequency $\frac{df}{dt}$ is influenced by the power imbalance ΔP and the system inertia constant, as described in equations (25)–(26).

$$\frac{2H}{f_0} \frac{df}{dt} = \overline{P}_m - \overline{P}_e = \Delta P. \quad (25)$$

$$H_{sys} = \frac{\sum_{i=1}^N VA_{base,i} H_i}{\sum_{i=1}^N VA_{base,i}} \quad (26)$$

To simplify comparisons and enhance computational convenience, this study represents system inertia through the rotational kinetic energy stored in synchronous generators, measured in $MW \cdot s$, as detailed in equation (27) [12].

$$E_{k,sys} = VA_{base,sys} H_{sys} = \sum_{i=1}^N VA_{base,i} H_i \quad (27)$$

- Rate of Change of Frequency

The Rate of Change of Frequency (RoCoF) indicates the severity of power balances and can be calculated using two frequency measurements taken consecutively: one before the disturbance and one after it occurs (28) [13].

$$\frac{df_i}{dt} = \frac{f_i(t^+) - f_i(t^-)}{t_i^+ - t_i^-}. \quad (28)$$

- Frequency Nadir

The Frequency Nadir (FN) refers to the minimum frequency value observed following a disturbance, such as the disconnection of a generating unit. This value is of great significance, as exceeding a predefined lower limit can activate Under-Frequency Load Shedding (UFLS) protection schemes.

III. CASE STUDY AND RESULTS

A. Study system

This study utilizes data from the Vietnamese power system at voltage levels of 220kV and 500kV, as outlined in PDP8. The grid model includes 1,724 buses, 2,369 transmission lines, and 868 transformers, as summarized in TABLE I.

TABLE I. VIETNAM POWER GRID DATA EXPECTED IN 2030

Number of buses	Number of transmission line		Number of transformers	
	220kV	500kV	2-winding	3-winding
640 generator buses				
218 500kV buses	1930	439	620	248
866 220kV buses	Total: 2369		868	

The regional distribution of generation sources in the Vietnamese power system is illustrated in Fig.3. The total installed capacity connected at voltage levels of 220kV and above is 147,266MW. Of this, RES accounts for 54,468 MW, which is approximately 36.99% of the total capacity [6]. It is

important to note that the capacity of RES connected at medium-voltage levels is deducted from the load data during the calculation. A key feature of the power generation system projected for 2030 is the significant integration of RES in the Northern region, with a total installed capacity of 15,158 MW. Additionally, there is a substantial amount of gas-fired generation, totaling 33,723 MW, distributed across all three regions of the country. The system also includes 5,310 MW of PSH and 730 MW of BESS. The inter-regional transmission capacity limits are set at 6,500 MW between the Northern and Central regions, and 6,600 MW between the Central and Southern regions.

According to PDP8, electricity demand is projected to grow at a rate of up to 10% per year from 2026 to 2030 [6]. In this study, we constructed the load profile for a representative week based on this forecasted growth rate and the typical load curve shape observed in Vietnam over recent years. The total load accounts for the capacity of distributed generators connected to voltage levels below 220 kV, which has been subtracted from the overall demand (Fig.4). Additionally, solar irradiance and wind speed data from ERA5 [14] have been collected and mapped to forecast renewable energy sources (RES) generation, following the methodology described in [15] (Fig.5).

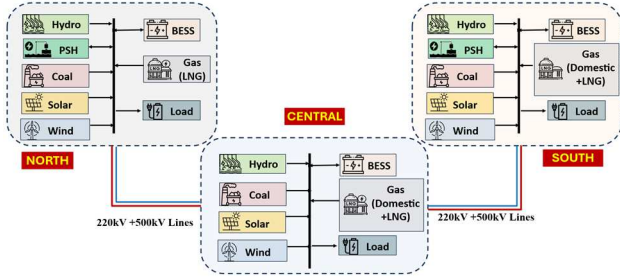


Fig. 3. Expected 2030 regional power generation mix in Vietnam.

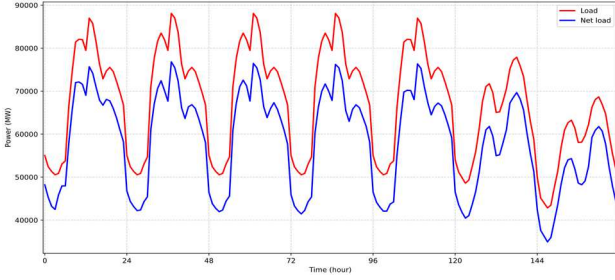


Fig. 4. Expected Load Profile for the Week of June 15–22, 2030.

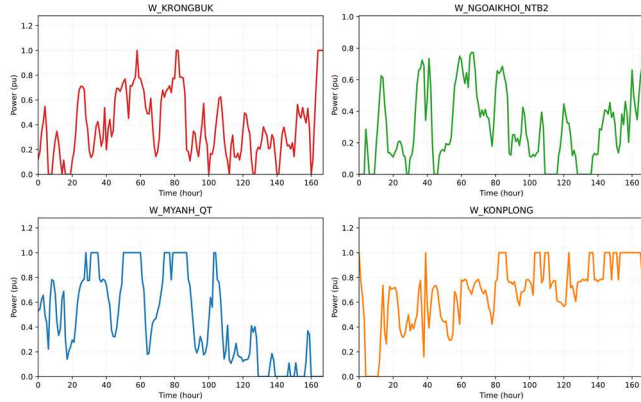


Fig. 5. Forecasted Power Output of Selected RES Plants.

B. Results and discussion

The framework in Section II was applied to the mentioned dataset to assess the impact of RES on the frequency stability of the Vietnamese power system in 2030. In this framework, operational scenarios were generated by solving the UC model, as presented in Section II, using the PyPSA library and the CPLEX solver. Then, these scenarios were used to simulate dynamic stability with the disconnection of the O Mon thermal unit (1030 MW) – the largest generating unit in the system. The simulations were run in PSS/E to analyze the system's Inertia, FN and RoCoF.

To evaluate the influence of RES on frequency stability in Vietnam's power system, this study focuses on the most critical operational periods of the year. The selected periods are the week of June 15, 2030 – June 22, 2030, and the week of February 15, 2030 – February 22, 2030. June is the time when the load consumption reaches its peak due to the hot weather in the North Region of Vietnam. In this period, hydro power will be heavily dispatched to meet the high load demand, since thermal plants will have less efficiency under high temperatures, and water reservoirs need to be prepared for the upcoming flood season. In contrast, February marks the beginning of the dry season, characterized by the lowest annual demand. Hydropower generators will be planned at a minimal level. Therefore, the analysis of operational characteristics and frequency stability is carried out under two scenarios:

- Case 1: From June 15, 2030 to June 22, 2030.
- Case 2: From February 15, 2030 to February 22, 2030.

1) Case 1: From June 15, 2030 to June 22, 2030

The simulation results for the week of June 15 – 22 are presented in Fig.6 – Fig.12. The optimal operational schedule of the system is illustrated in Fig.6. Based on those scenarios, dynamic stability simulations were performed under a disturbance of tripping the largest generating unit. The Frequency Inertia and RoCoF values are shown in Fig.7 and Fig.10. In this study, RoCoF was calculated over a 500ms window from the time of the trip event (Fig.9), about the methodology adopted in study [13].

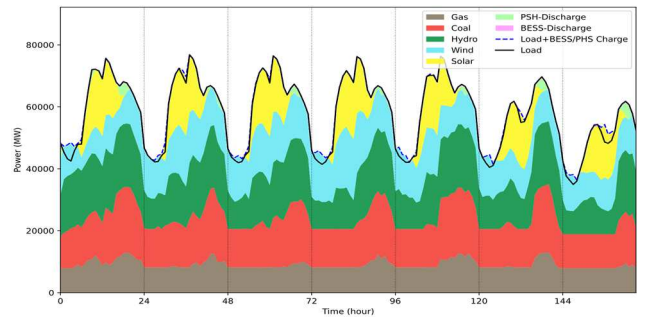


Fig. 6. Case 1: Operation Schedule of generation units.

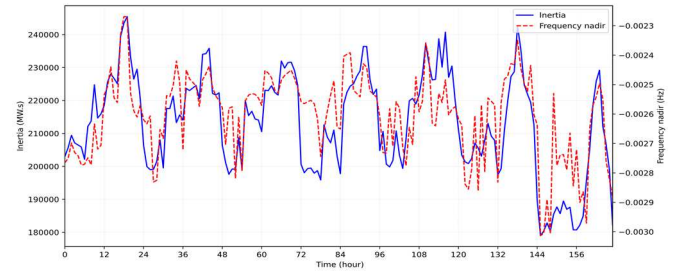


Fig. 7. Case 1: Frequency Nadir and System Inertia.

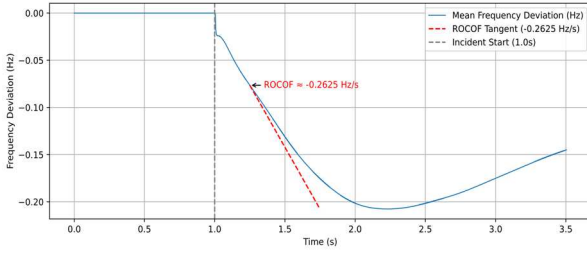


Fig. 8. System RoCoF at 0h, June 15, 2030.

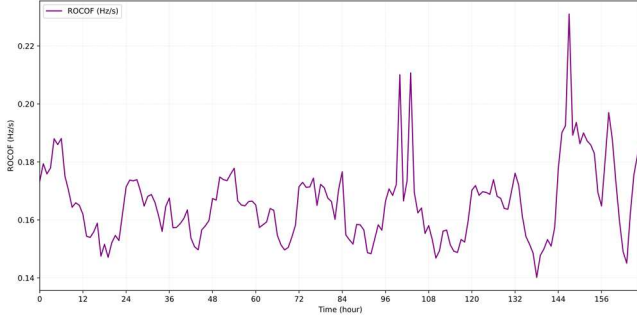


Fig. 9. Case 1: System RoCoF.

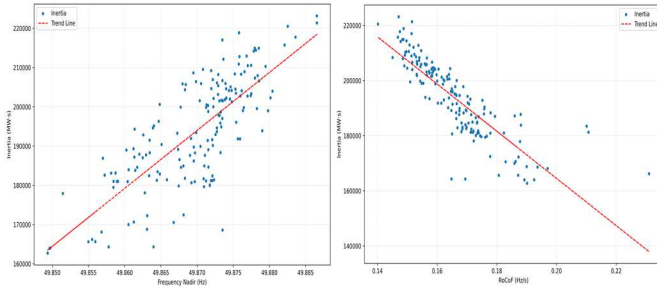


Fig. 10. Case 1: Relationship between indicators.
(a) Frequency Nadir and Inertia; (b) RoCoF and Inertia

From Fig.6, RES consistently accounts for a high in operation, especially in the 60th hour of the week, the share of RES was 37.64%. Almost the entire week from June 15 to June 22, the system always has to dispatch a large amount of power from conventional generators, especially hydropower, to meet the elevated load demand. This is partly due to transmission limits that prevent RES in the central region from operating at their full capacity. Consequently, the system maintains a high inertia of around 175502 MW.s and 245821 MW.s. RoCoF value ranges from 0.14 Hz/s to 0.24 Hz/s (Fig.10), while Frequency Nadir varies between 49.85Hz and 49.883Hz (Fig.5). It is be noted that the FN value should higher than reality because in this study the governor models were built based on suggested templates and there is no deadband, therefore they have a faster reaction time compared to actual governor.

It is evident that FN tends to decrease and RoCoF increases during nighttime (Fig.10), especially from the 144th hour to the 147th hour, when load demand drops significantly and wind turbines still remain high. This made the system's inertia low during these periods. Conversely, during peak hours, particularly in the evening when a large number of synchronous generators are dispatched, inertia and FN have a rising trend. Moreover, the optimized scheduling of BESS and PSH by charging/pumping during off-peak hours and generating during peak hours (Fig.6) contributes to improved frequency response during these times. In addition, the relationship between system inertia and FN in Fig.8 shows

that in most scenarios, as rotational kinetic energy increases, FN also increases and vice versa. This reflects the physical relationship between these two quantities. In contrast, there is an inverse relationship between inertia and RoCoF as depicted in Fig.11.

In summary, during June, the system benefits from high inertia due to the extensive dispatch of conventional generators, especially hydro power. Thus, the system always maintained a high level of inertia, the FN and RoCoF indices were all in the good value range. The inertia, frequency nadir and RoCoF indices of the system all correctly reflected the physical relationship.

2) Case 2: From February 15, 2030, to February 22, 2030

The results for the week of February 15 – 22 are presented in Fig.11 – Fig.13. Fig.11 shows the system operation plan while Fig.12 and Fig.13 illustrate the inertia, FN, and RoCoF indicators for the same period. The results show that the inertia during this period reached values from 143200MW.s to 211103MW.s, FN reached values from 49.82 Hz to 49.87 Hz and RoCoF reached values from 0.14Hz/s to 0.24Hz/s (Fig.12, Fig.13).

This shows that February is the time when RoCoF tends to increase. Thus, February will be the time that needs more attention in operation due to the high proportion of RES, leading to a low FN and a high RoCoF tendency.

In February, when the load was low and the dispatch of hydropower was reduced, the proportion of RES in operation was higher than in June. This can be observed in Fig.8: at hour 84 of the week of February 15 - 22, RES share reached 52.59%, while in the week of June 15 - 22, the maximum RES share was only 37.64%. Therefore, in February, both system inertia and FN are lower than in June. This can be seen when looking at the inertia and FN values in TABLE II and Fig.14, where the data points of Case 2 tend to be lower than those of Case 1. With RoCoF, it can be seen that both weeks have the same value range from 0.14Hz/s – 0.24Hz/s (TABLE II), but in general, most of the data points of Case 2 are lower than those of Case 1. This shows that February is the time when RoCoF tends to increase. Therefore, the high RES penetration in February, which leads to a low FN and high RoCoF, calls for greater operational attention and the consideration of measures to enhance frequency stability, such as requiring virtual inertia support from RES and integrating frequency stability constraints into the UC problem.

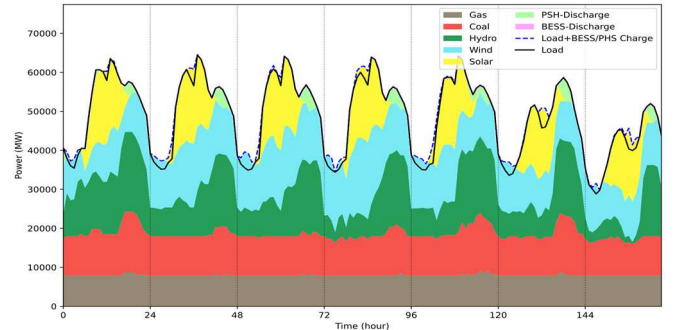


Fig. 11. Case 2: Operation Schedule of generation units.

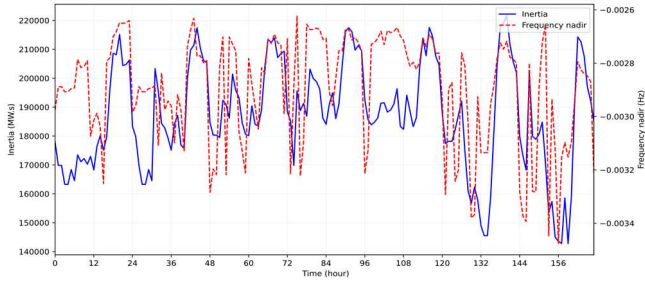


Fig. 12. Case 2: Frequency Nadir and System Inertia.

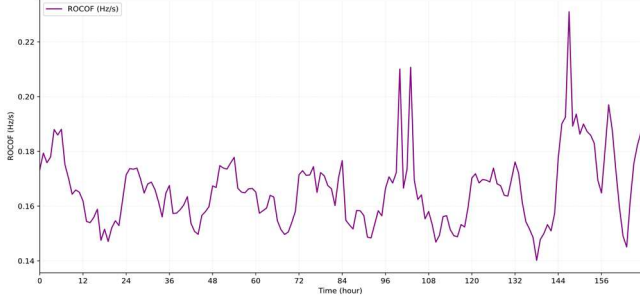


Fig. 13. Case 2: System RoCoF.

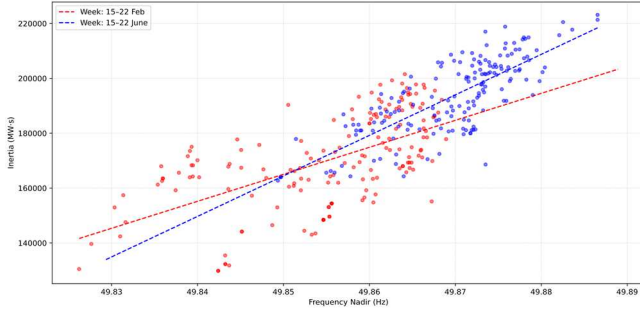


Fig. 14. Comparison of System Inertia and FN between Case 1 and Case 2.

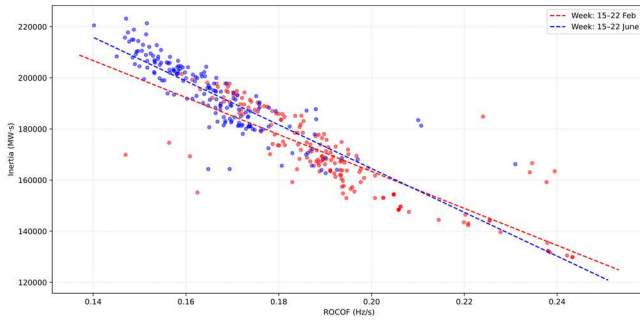


Fig. 15. Comparison of System Inertia and RoCoF between Case 1 and Case 2.

TABLE II. COMPARISON BETWEEN CASE 1 AND CASE 2.

Week	Share of RES (%)	Inertia (MW.s)	Frequency Nadir (Hz)	RoCoF (Hz/s)
15/2 – 22/2	52.59	143200 – 211103	49.85 – 49.88	0.14 – 0.24
15/6 – 22/6	37.64	175502 – 245821	49.82 – 49.87	0.14 – 0.24

IV. CONCLUSION

This paper proposed an integrated process to assess the impact of RES sources on frequency stability based on optimal operation scenarios. The framework could also be applied to the Vietnamese power system as projected in the PDP8 by 2030. Simulation results for two scenarios

corresponding to two critical operating periods revealed: In June, the Vietnamese power system has high inertia due to the dispatch of a significant amount of conventional generators, especially hydro power. As a result, favorable values for frequency nadir và RoCoF were observed. Conversely, in February, RES accounts for 52.59% in operation, so the system has a lower inertia and frequency nadir, but RoCoF tends to be higher. Therefore, the operating periods in February pose higher risks to frequency stability compared to those in June. To enhance frequency stability, several measures should be considered, such as requiring synthetic inertia support from RES plants and incorporating frequency stability constraints into the UC problem, in order to balance between minimizing operating costs and ensuring secure system operation.

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